Matrix
identity
transpose
determinant
inverse

Definition

An **identity matrix** is a square matrix with:

- **1s on the main diagonal**
- **0s everywhere else**

Examples

2×2 identity matrix

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3×3 identity matrix

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Key Property

For any matrix A:

$$AI = IA = A$$

It behaves like the number **1** in multiplication.

$$\overset{A}{\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}} \overset{I}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} 2\times1 + 3\times0 & 2\times0 + 3\times1 \\ 5\times1 + 7\times0 & 5\times0 + 7\times1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} = A$$

$$\overset{I}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \overset{A}{\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}} = \begin{bmatrix} 1\times2 + 0\times5 & 1\times3 + 0\times7 \\ 0\times2 + 1\times5 & 0\times3 + 1\times7 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} = A$$

The **transpose** of a matrix swaps rows and columns.

Notation:

$$A^T$$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Transpose:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Important Facts

- Rows → columns
- Columns → rows
- Size changes from $m \times n$ to $n \times m$

The **determinant** is a single number computed from a **square matrix**.

Notation:

$$\det(A) \quad \text{or} \quad |A|$$

For a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Determinant:

$$|A| = ad - bc$$

Example

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$|A| = (2)(4) - (1)(3) = 8 - 3 = 5$$

The **inverse** of a matrix A:

Notation:

$$A^{-1}$$

Fundamental Property

$$AA^{-1} = A^{-1}A = I$$

Multiplying a matrix by its inverse gives the **identity matrix.**

Important Condition

A matrix has an inverse **only if**:

$$\det(A) \neq 0$$

Inverse of a 2×2 Matrix

If:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

then:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Determinant:

$$(2)(3) - (1)(1) = 5$$

Inverse:

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

Verify

$$A A^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{5} \cdot \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Application

Solve:

$$2x + y = 5$$
$$x + 3y = 6$$

Write in Matrix Form

$$AX = B$$

$$\overset{A}{\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}} \overset{X}{\begin{bmatrix} x \\ y \end{bmatrix}} = \overset{B}{\begin{bmatrix} 5 \\ 6 \end{bmatrix}}$$

Application

Solve Using the Inverse

$$X = A^{-1}B$$

After find inverse of A

$$X = \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$X = \begin{bmatrix} 9/5 \\ 7/5 \end{bmatrix}$$

$$x = 1.8, \quad y = 1.4$$

```
A = np.array([[2, 1],
              [3, 5]])

det_A = 2*5 - 1*3 = 7

A = np.array([[2, 1, 0],
              [0, 1, -1],
              [0, 2, 4]])
det_A = 12
```